AI Monoculture in the Stock Market: Do LLM-Generated Trading Agents Converge?

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Abstract

Increasingly, people are using large language models (LLMs) to design trading strategies, which could lead many traders to hold and trade the same stocks, raising the question: do trading agents independently generated by the same LLM converge more than chance, and does mixing providers reduce it? To answer this question, Claude Sonnet 5 from Anthropic, Gemini 3.6 Flash from Google, and GPT-5.6 Terra from OpenAI were used to generate 30 trading agents each, from the same open prompt. The agents were run on 30-minute data for 486 S&P 500 stocks over 2025, and were compared with random and popular comparison agents using holdings overlap, trading direction, and crowding. The main results showed that same-model agents overlapped more than different-model agents, and all AI agents overlapped significantly more than chance (e.g. within-model holdings overlap, measured by Jaccard similarity, of 0.189 vs 0.012 for random agents). GPT showed the strongest model-specific style, Claude a partial one, and Gemini none. Mixing LLMs removed only about a third of the excess overlap (diversification floor of 66%, 95% confidence interval 55.0–76.2%) and almost none of the one-sided trading. Hence, AI-generated trading strategies share an AI-wide monoculture that switching LLMs largely fails to remove, which is a potential precondition for crowded sell-offs in the stock market.

1. Introduction

As large language models (LLMs) evolve and more and more people use AI models like ChatGPT, Claude, and Gemini to design trading strategies and write trading code, many traders may get similar strategies when using the same AI, thus opening a possibility in which they may hold and trade the same stocks at the same time. In finance, crowded positions are a known risk, because when traders holding the same positions sell together, prices can fall sharply (e.g. the August 2007 quant crisis; Khandani & Lo, 2011). Regulators have begun raising this concern about AI specifically as well (FSB, 2024).

Prior work shows correlated behaviour among LLM agents in simulations (Ross et al., 2026) and warns about "algorithmic monoculture" in general (Kleinberg & Raghavan, 2021). However, to my knowledge, no study has directly measured how alike these independently generated AI trading programs are on real market data, or tested whether using agents from different LLM providers reduces that similarity.

Based on past research and the current state of AI usage involving the stock market, the formulated research question became: Do trading agents independently generated by the same LLM converge more than chance, and does mixing providers reduce it? In order to answer this question, 30 agents each from Claude, Gemini, and GPT were generated using the same open prompt and run on 30-minute data for 486 S&P 500 stocks, scored over 2025, with overlap, trading direction and crowding compared against comparison agents (random and popular).

The main findings were that the agents generated from the same model overlapped more than agents from different models, and all AI agents overlapped far more than chance. GPT

agents showed the strongest model-specific style, Claude agents a partial one, and Gemini agents none. Mixing providers removed only about a third of the excess overlap (a diversification floor of 66%) and almost none of the one-sided trading seen in these agents' behaviours.

This study makes three contributions: (1) to my knowledge, the first measurement of convergence among independently generated AI trading programs on real market data; (2) a direct test of whether provider diversity reduces it; and (3) evidence of two layers of monoculture, an AI-wide one and a model-specific one. Section 2 describes the methods, Section 3 the results, Section 4 discusses their implications, Section 5 the limitations, and Section 6 concludes.

2. Methodology

2.1 Overview

I used three large language models (LLMs), Claude Sonnet 5 from Anthropic, Gemini 3.6 Flash from Google and GPT-5.6 Terra from OpenAI, to generate 30 trading agents each, 90 in total. Each agent was generated from its own single request using the same prompt, which asked the model to design its own trading strategy from scratch and write it as a program. Each program was then run on historical prices of S&P 500 stocks as listed in January 2024, with 2024 used as a warm-up period and 2025 as the scored period. Because some stocks had no trades in certain 30-minute intervals, I kept only stocks that had a price in at least 99% of the bars, which left 486 of the original 503. Every agent received the same data: 30-minute price and volume bars for all 486 stocks. I measured holdings overlap, trading direction and crowding, both among agents from the same model and among agents from different models, as well as how these changed over the course of 2025 and between calm and volatile market conditions. Finally, I

compared all of these results against two kinds of comparison agents: one that picked stocks entirely at random, and one weighted towards the stocks the AI agents bought most often.

2.2 The Prompt

The prompt given to each model was split into two parts: an open request (the free half) and the technical rules of the platform (the fixed half). The open request read: "I want a trading agent. Design the approach yourself — how it picks what to hold, when to move, how much to commit, all of it. Write the code."

The fixed half required every agent to be written in Python 3.11 with pandas and numpy available, and with no network or file access. It defined a function, init(cash), called once before the first bar with a starting balance of $100,000 for every agent. A second function, on_bar(history), was called once per 30-minute bar with the full price and volume history up to the current bar, and returned a list of orders, each giving the stock, the direction (buy or sell) and the number of shares. Agents were not told the bar length, the market, or which index the stocks came from.

Because a bar's closing price is only known once its 30-minute interval has ended, every order was filled at the next bar's opening price. Trading carried no commissions, slippage or market impact, so execution was identical for every agent and differences in behaviour came from strategy rather than cost assumptions. Agents received no fill reports or account statements at any point in the run, so each had to track its own positions and cash balance as part of its strategy. Orders were filled partially when necessary: an agent received as many shares as its cash or holdings allowed, and the remainder was discarded rather than carried forward, so no agent could spend cash it did not have or sell shares it did not hold. An order filled for zero

shares if the stock had no price on the execution bar. Agents could not short or borrow, so holdings were never negative and overlap measures such as Jaccard similarity and cosine similarity were well-defined. Each agent's code had to complete the full run within 60 minutes or it was terminated. The full prompt is given in Appendix A.

2.3 Agent Generation

The prompt was sent word for word as a single message, with no system prompt, tools or past conversation, one API call per agent, in August 2026. Each agent therefore started from a blank slate, and none saw another agent’s output. The code was taken from each response as its longest Python code block, and the full response was saved. Every agent was kept exactly as generated, with no code fixed or replaced, since replacing failed agents would keep only the ones that work and the population would no longer show what each model actually produces. Table 1 lists the settings; all three models ran at each provider’s default sampling, so the settings are comparable but not numerically identical.

Table 1. Generation settings for each model.

	Model ID	**Sampling**	**Max output tokens**
Claude	claude-sonnet-5	temperature 1.0 (default)	32,000
Gemini	gemini-3.6-flash	temperature 1.0 (default)	40,000 (thinking tokens count towards this)
GPT	gpt-5.6-terra	no temperature (reasoning models don't accept one); reasoning effort "medium" (default)	32,000 (reasoning tokens count towards this)

2.4 Market Data

Market data was sourced from Alpaca's historical data API using the SIP feed, which combines trades from all US exchanges. Data covered January 2024 to December 2025; 2024 served as a warm-up and only 2025 was scored. The data came as bars, each containing a ticker's open, high, low and close prices and its volume over a 30-minute interval. The data contained regular sessions only, from 9:30 to 16:00 ET, or 13 bars per day. The prices were adjusted for splits and dividends. The initial simulated market consisted of the 503 tickers in the S&P 500 as of January 2024, held fixed for the whole period so that stock selection did not depend on later index changes. However, since only 22 stocks had a bar at every timestamp, tickers that had a bar in at least 99% of the timestamps were kept, leaving 486. In total, the scored period contained 3,250 bars across 250 trading days.

2.5 Running the Agents

A separate program ran each agent, rather than the agent running itself. It called init(cash) once, then on_bar(history) once per bar in time order, from the first 2024 bar to the last 2025 bar, and it made sure to only give each agent data up to and including the current bar to prevent the agents from seeing future prices. Every order was saved with its bar, stock, direction and size, and malformed orders were dropped and logged as warnings. No agent reached the time limit, and for crashed runs, the orders placed before the crash were kept as part of the result. The

program only recorded the orders; the fills, holdings and cash were worked out afterwards (Section 2.6).

2.6 Rebuilding Holdings

The orders recorded by the program in Section 2.5 were replayed against the price data with the fill rules from Section 2.2, starting from the first 2024 bar so that positions carried into 2025 correctly, to find what each agent actually held. Though this is not specified in the prompt, the sell orders were filled first so the agent's cash could fund the same-bar buys, and buys were then filled in the order the agent listed them. Holdings were valued at each bar's closing price, and stocks with no price kept their last known price. For every agent and bar, shares held, cash and the value of each position were recorded for computation. Each agent's fill rate was the share of its 2025 orders that filled at least one share.

2.7 Measures

All measures used the scored bars (2025) only, computed from the holdings rebuilt in Section 2.6. Every measure was also computed for the two groups of synthetic comparison agents described in Section 2.8, to show what each value looks like by chance. Agents that held nothing for more than 95% of scored bars were excluded from the overlap measures. This removed 3 agents, gemini_002, gemini_026 and gpt_004, since their portfolios had nothing to compare.

First, the average number of stocks held, median holding period, number of orders and fill rate were measured for each agent to describe what the agents are like before comparing them.

Holdings overlap was computed for every pair of agents at every bar, then averaged over bars. These include the Jaccard similarity of the stocks held by each pair, weighted Jaccard, which takes into account how much of each portfolio sits in the shared stocks, and cosine similarity, which measures how similarly the pair split their money across stocks, with the largest positions counting most. Bars where both agents held nothing were skipped, since overlap is undefined. Furthermore, pairs were grouped as either within-model (same model) or cross-model (different models, AI agents only) depending on which model generated the agents. These groupings were used to measure how much more alike agents from the same model are than agents from different models, or the convergence ratio. The diversification floor measured how much of a model's above-chance overlap remained when its agents were paired with agents from other models instead, where 100% means mixing models does not reduce overlap at all and 0% means it brings overlap down to the level of random agents.

For each stock in each bar, the flow imbalance, or how one-sided the agents' trading was, was calculated. It specifically measured how the shares bought by agents trading the same stock at the same time compared with the shares sold, based on the orders the agents placed. Stock-bars traded by only one agent were dropped, since a lone trader is one-sided by default, and the rest were weighted by how many agents traded that stock. This was measured for each model alone (Claude, Gemini and GPT), for mixed groups of 10 agents per model (10+10+10, re-drawn 200 times), and for all 90 AI agents pooled, which was used only as a description since the larger group size alone lowers the score.

Crowding was measured as the effective number of stocks: how many stocks a group's combined holdings are really spread across, counted both by how many agents hold each stock and by how much money is in it. It was computed for each model alone and for a mixed group of 10 agents per model (10+10+10, re-drawn 200 times). Because whole-group measures change with group size, the mixed group was made the same size as a single-model group and re-drawn 200 times, so differences reflect mixing models rather than adding agents or the luck of one draw.

Finally, every measure was also computed bar by bar, so it could be tracked across 2025 and compared between calm and volatile market conditions (Section 2.9).

2.8 Comparison Agents

Although I wanted to compare the behaviour of the agents with actual human traders, no public data records individual human traders' holdings at 30-minute resolution. Hence, comparison agents were necessary to objectively compare the results of the AI agents to chance and to a shared preference for popular stocks.

Two groups of 30 comparison agents, random and popular, were generated and went through the same replay and measures as the AI agents. Each comparison agent copied the portfolio size and holding period of a randomly chosen AI agent, since larger portfolios overlap more by chance. Every holding period, it sold everything and bought a fresh set of stocks, splitting its cash equally among them. Random agents were designed to pick from the 486 tickers by pure chance, thus creating a pure-chance baseline to evaluate against. Popular agents picked stocks in proportion to how often the AI agents bought them in 2025, so that popular stocks got

picked more. By doing so, this separates two explanations for the overlap: the AI agents thinking alike, or simply all favouring the same popular stocks.

The comparison agents were not given the 2024 data as warm-up and started trading at the beginning of 2025, since they need no history.

2.9 Statistical Analysis

A permutation test was run in order to answer the question of whether a model's within-model overlap was higher than when grouping agents at random regardless of the AI model that generated it. A permutation test was necessary and not a t-test because every agent appears in many pairs, therefore a standard test would be overconfident as it assumes each data point is independent from each other.

The test was done by picking, at random, a group the same size as the model's own group (30 for Claude, 28 for Gemini, 29 for GPT) from the 87 AI agents, excluding the comparison agents. For flow imbalance, all 90 AI agents were used, since agents without orders contribute nothing. The chosen agents were measured by the overlap of all pairs among those agents. This process was repeated 5,000 times.

The results were compared to the overlap measured by the actual groupings of AI agents generated within the same model. If the real overlap was above the 95th percentile (set before the results were seen) of the 5,000 shuffled values, the model passed.

The share of these random shuffles that matched or beat the actual values was taken as the p-value. It was calculated as: (number of random shuffles that matched or beat the actual value + 1) / (5,000 + 1). +1 was added to both the denominator and the numerator in order to make sure the p-value is never 0, as that would mean it is impossible to randomly match or beat the real value, which 5,000 shuffles cannot prove. In addition, a z-score was reported, z = (observed − shuffled mean) / shuffled standard deviation, expressing how many standard deviations the observed value lies above the average shuffled value. The z-score is used to describe the size of the departure from chance. The significance was judged from the empirical p-value rather than a normal approximation, since the shuffled distribution need not be symmetric. These were calculated for each model separately, for Jaccard, cosine and flow imbalance, and for all three models pooled, for Jaccard and cosine. The test was also repeated with the comparison agents included in the pool, to test each model against chance.

To quantify the sampling uncertainty of the diversification floor (Section 2.7), that is, how much the estimate would vary had different agents been generated, a 95% bootstrap confidence interval (CI) was constructed. This was done by resampling each model's agents at random with replacement, keeping the same group size, and recomputing the floor with the re-drawn set 2,000 times. The middle 95% of the 2,000 recomputed floors gave the interval. Agents rather than pairs were resampled, since pairs share agents.

In order to test whether agent crowding intensifies under market stress, all bars of 2025 were labelled calm, normal, or stressed. This was done by creating an equal-weighted index of all 486 stocks from bar-level closing prices, normalised to each stock's first price in the scored period. The daily index returns were computed from the last bar of each trading day, and the realised volatility was taken as the standard deviation of those returns over a rolling five-day

window. The trading days were split into terciles of realised volatility and labelled calm for the lowest third, normal for the middle, and stressed for the highest third. Every bar took its day's label, and since the labels came only from market data, every agent was classified identically. Holdings overlap and flow imbalance were then averaged within each regime and compared between calm and stressed periods, alongside the random comparison agents.

In order to check whether a failed test meant that a model's agents truly had no distinct style, or simply that there were not enough agents to detect it, a power analysis was run. This was done by creating groups of simulated agents with a known amount of overlap, where each agent picked part of its stocks from a shared set of stocks given to its group. The same permutation test was then run on these simulated agents many times with 30 agents per group, to see how often the test was able to detect the overlap. Agents were also grouped by the similarity of their holdings using hierarchical clustering (average linkage on 1 − Jaccard).

3. Results

3.1 Final Sample

In total, 90 AI agents (30 per model) plus 60 comparison agents (30 random, 30 popular), or 150 agents, were generated. Of the 90 AI agents, 88 ran the full period without error. Two GPT agents stopped with a runtime error. gpt_004 crashed before buying anything, and gpt_011 crashed after buying 8 stocks, and those positions stayed in place for the rest of the run. Two Gemini agents (gemini_002, gemini_026) ran without errors but never placed an order and held cash throughout the run. The overlaps were measured with 87 AI agents (30 Claude, 28 Gemini, 29 GPT), after excluding the 3 that never held anything (Section 2.7), while the flow imbalance was measured for all 90 AI agents (Table 2).

Table 2. Agents generated and included in the analysis, by model.

Model	Generated	Ran without error	Crashed	Never traded	Included in overlap measures	Included in flow imbalance
Claude	30	30	0	0	30	30
Gemini	30	30	0	2	28	30
GPT	30	28	2	0	29	30
Total	**90**	**88**	**2**	**2**	**87**	**90**

Note. One crashed GPT agent (gpt_011) held 8 stocks before stopping and was kept in the overlap measures; the other (gpt_004) held nothing and was excluded. Agents that never traded held cash throughout. The 60 comparison agents (30 random, 30 popular) are not shown.

3.2 Agent Behaviour

When measuring the portfolio size for all AI agents, Claude agents held the most stocks on average, 11.8 stocks with a median of 10.0. Gemini agents held 7.4 stocks on average, with a 6.9 median. GPT agents held 6.6 stocks on average and 6.1 for the median. In total, the average of the AI agents was 8.6 stocks (Table 3).

The median holding period (the number of bars from buying a stock to fully selling it) was 10 bars (5 hours) for Claude and GPT agents, but only 2 bars (1 hour) for Gemini agents, showing that Gemini agents traded much faster than the other agents.

The median fill rate came out as 99.8% for Claude, 99.7% for Gemini, and 99.8% for GPT agents, showing that the AI agents tracked their own cash and holdings accurately and consistently without fill reports. However, one Claude agent (agent_003) filled only 44.1% of its orders, mostly because it tried to sell stocks it didn't hold. This is the only agent with real tracking failure, as every other AI agent filled at least 98.1%.

The random agents of the comparison agents held 12.5 stocks on average, and popular agents held 11.7, both of which are higher than the AI average of 8.6. Since bigger portfolios overlap more by chance, the chance baseline used to measure the actual agents' overlap sits slightly high, therefore the gaps to chance reported below are conservative.

The fill rates of the comparison agents, popular and random, were lower, with a median of 87.2% for random agents and 91.8% for popular agents.

Table 3. Portfolio size, holding period and fill rate, by group.

Group	Mean stocks held	Median stocks held	Median holding period (bars)	Median fill rate	Min fill rate
Claude	11.8	10	10	99.80%	44.10%
Gemini	7.4	6.9	2	99.70%	98.10%
GPT	6.6	6.1	10	99.80%	99.30%

Popular	11.7	9	5	91.80%	80.20%
Random	12.5	11	5	87.20%	77.10%

Note. One bar = 30 minutes. Stocks held are averaged over agents with any holdings. Fill rates exclude agents with no 2025 orders (2 Gemini, 2 GPT).

3.3 Holdings Overlap

The main finding in holdings overlap was that the agents from the same model held more of the same stocks than agents from different models, as the average Jaccard was 0.189 within models compared to 0.129 across models (Table 4).

By model, GPT overlapped the most at 0.226 average Jaccard, then Claude at 0.194, then Gemini at 0.143. The Jaccards derived from the models were far higher than those of the comparison agents. Pairs of random agents averaged 0.012 and popular agents' pairs 0.015. The within-model overlap was about 19 times random for GPT, 16 times for Claude and 12 times for Gemini. Even the cross-model AI pairs were about 11 times random agents and 9 times popular agents. When comparing between models, Claude and GPT agents resembled each other more, with an average Jaccard of 0.153, than either resembled Gemini agents, as the average Jaccard was 0.117 for Claude–Gemini pairs and 0.115 for GPT–Gemini pairs.

When money was taken into account, using cosine similarity and weighted Jaccard, the results were similar. The cosine similarity for GPT was 0.349, Claude 0.257, Gemini 0.207 when

calculated within models and 0.195 for cross models, while random agent pairs were at 0.022 and popular agents at 0.028. GPT agents came in first for the weighted Jaccard at 0.188, second was Claude with 0.146, and last was Gemini at 0.107. As for the comparison agents, popular agents had a 0.011 weighted Jaccard while random agents had 0.009.

The convergence ratio (within-model average Jaccard / cross-model average Jaccard) reached 1.68 for GPT agents, 1.43 for Claude agents, and 1.23 for Gemini agents, indicating that each model's agents resembled each other more than they resembled agents from other models, most strongly for GPT agents.

When looking at the Jaccard of individual pairs of agents, 8 Claude pairs had Jaccard above 0.9, while no GPT or Gemini pair did. As a result, Claude's average is pulled up by a handful of near-copies, while GPT and Gemini's overlap is spread more evenly.

Table 4. Mean holdings overlap by pair group.

Pair group	Pairs	Jaccard	Weighted Jaccard	Cosine
Claude–Claude	435	0.194	0.146	0.257
Gemini–Gemini	378	0.143	0.107	0.207
GPT–GPT	406	0.226	0.188	0.349
Claude–Gemini	840	0.117	–	0.161
Claude–GPT	870	0.153	–	0.25
Gemini–GPT	812	0.115	–	0.171
All within-model	1,219	0.189	–	0.272
All cross-model	2,522	0.129	–	0.195
Popular–Popular	435	0.015	0.011	0.028

Random–Random	435	0.012	0.009	0.022

Note. Values are averaged over all 2025 bars and all pairs in each group. Weighted Jaccard was not computed separately for cross-model pairs.

3.4 Significance

In the pooled result totaling all three models, same-model pairs overlapped far more than random regroupings of AI agents: Jaccard z-score = 15.14, cosine z-score = 15.53, p-value for both = 0.0002 (lowest value possible), meaning none of the 5,000 shuffles matched or beat the real value (Table 5).

Specifically, GPT agents passed all three measures (Jaccard, cosine and flow imbalance) using the gate set in Section 2.9, with a Jaccard z-score of 4.57 and p-value of 0.0002, cosine z-score of 6.05 and p-value of 0.0002, and flow imbalance z-score of 2.04 and p-value of 0.010. This reveals that GPT agents held the same stocks and put money in the same big bets, while trading in the same direction more than any random group of AI agents.

As for the Claude agents, they passed on holdings overlap but not when the amount of money was accounted for or on trading direction. The Jaccard z-score was 2.71 and p-value 0.007, therefore passing the gate (Section 2.9). However, cosine failed by the smallest possible margin, as the observed cosine value was 0.2565 while the gate was set at 0.2566. The z-score of cosine was 1.69 and the p-value was 0.051. Flow imbalance also failed, with a z-score of 1.42 and p-value of 0.080.

Gemini agents failed every test. Their Jaccard p-value was 0.610 and z-score −0.32, cosine p-value 0.732 and z-score −0.65, and flow imbalance p-value 0.992 and z-score −2.59. These numbers show that Gemini agents were no more alike than a random group of AI agents.

When comparison agents were included in the pool, all three models passed on Jaccard and cosine, including Gemini. GPT's Jaccard p-value and z-score were 0.0002 and 10.36, respectively. Claude's Jaccard p-value and z-score were 0.0002 and 8.65, respectively. Gemini's Jaccard p-value and z-score were 0.0002 and 5.33, respectively. For cosine, the p-value was 0.0002 for all three, with z-scores of 11.76 for GPT, 7.39 for Claude and 5.26 for Gemini. The random and popular agents failed, with z-scores between −2.84 and −3.10.

The power analysis showed that, with 30 agents per group, the permutation test was able to reliably detect overlap about 90 times smaller than the effect observed. However, since the simulated agents were more uniform than the real ones, this overstates how sensitive the test is. Gemini's within-model overlap (0.143) was also lower than the average of the shuffled groups (0.149), therefore Gemini's failure was due to its agents not having a distinct style rather than there not being enough agents.

Table 5. Permutation test results, AI agents only (5,000 shuffles).

Measure	Group	Observed	Shuffled mean	Gate (95th pct)	p-value	z-score	Result
Jaccard	Claude	0.194	0.148	0.178	0.007	2.71	Pass
	Gemini	0.143	0.149	0.178	0.610	−0.32	Fail
	GPT	0.226	0.149	0.178	0.0002	4.57	Pass
	Pooled	0.189	0.149	0.153	0.0002	15.14	Pass
Cosine	Claude	0.2565	0.220	0.2566	0.051	1.69	Fail
	Gemini	0.207	0.220	0.257	0.732	−0.65	Fail

	GPT	0.349	0.220	0.257	0.0002	6.05	Pass
	Pooled	0.272	0.220	0.226	0.0002	15.53	Pass
Flow imbalance	Claude	0.753	0.718	0.759	0.080	1.42	Fail
	Gemini	0.655	0.718	0.755	0.992	−2.59	Fail
	GPT	0.773	0.720	0.759	0.010	2.04	Pass

Note. Pass means the observed value exceeded the 95th percentile of 5,000 shuffles of model labels among AI agents. A p-value of 0.0002 is the minimum possible (no shuffle matched). Flow imbalance has no pooled test because it is measured on whole groups. Claude's cosine is shown to four decimals because it missed the gate by 0.0001.

3.5 Trading Direction (Flow Imbalance)

Section 3.4 explained whether each model's flow imbalance exceeded that of random groups of AI agents; this section compares the values with the comparison agents and examines mixed groups.

Flow imbalance was measured for all stock-bars where at least two agents traded the same stock at the same time (Section 2.7), roughly 55% of stock-bars, while the other 45% had only one trader and were dropped. The number of bars in which at least two agents from the same model traded the same stock was 818 for GPT, 1,404 for Claude and 3,250 for Gemini.

The flow imbalance was 0.773 for GPT, 0.753 for Claude and 0.655 for Gemini, against 0.591 for popular agents and 0.579 for random agents, the level of one-sidedness expected by chance.

When flow imbalance was measured for size-matched mixed groups (10 agents from each model, re-drawn 200 times), it scored 0.717 on average, with a 95% range of 0.671 to

0.765. The average single-model group scored 0.727. Measured against random agents, the mixed groups kept about 93% of the extra one-sidedness of a single-model group (Table 6).

All 90 AI agents together scored a 0.648 flow imbalance, reported as a description only, since a group three times larger scores lower because of its size.

Table 6. Flow imbalance of mixed and comparison groups.

Group	Agents	Flow imbalance
Average single-model group	30	0.727
Mixed (10 per model, 200 draws)	30	0.717 (0.671–0.765)
Popular	30	0.591
Random	30	0.579
All AI agents (descriptive)	90	0.648

Note. 1 = all agents traded the same direction; 0 = buys and sells fully offset. Values for individual models are in Table 5. The mixed group shows the mean and 95% range over 200 draws.

3.6 Crowding (Effective Number of Stocks)

By headcount, the effective number of stocks came out at 18.6 for GPT, 30.9 for Gemini, 45.7 for Claude, 37.2 for the mixed grouping, 180.3 for popular, and 215.6 for random agents. As a whole, all 87 active agents had a crowding of 39.6 (Table 7).

The crowding measured, when weighted by money, showed that GPT agents spread across only 16.8 stocks, 26.5 stocks for Gemini, and 28.7 for Claude, or 24.0 on average out of a possible 486 stocks to choose from.

On the other hand, random agents of the comparison agents spread across 177.6 stocks and popular agents across 138.6 stocks, roughly 7.4 and 5.8 times more than the average single-model AI crowd.

When the grouping is mixed for the AI agents, where 10 agents per model were chosen to form a group of 30 randomly selected agents (200 draws), crowding spread across 28.1 stocks, about 17% wider than the average single-model crowd (24.0). Moreover, mixing mainly diluted GPT's concentration, as it widened about 1.7 times from 16.8, while Claude's and Gemini's barely changed.

Table 7. Effective number of stocks held by each crowd (2025 average).

Crowd	Agents	By headcount	By money
GPT	29	18.6	16.8

Gemini	28	30.9	26.5
Claude	30	45.7	28.7
Average single model	–	31.7	24.0
Mixed (10 per model, 200 draws)	30	37.2 (27.2–52.1)	28.1 (24.0–32.8)
Popular	30	180.3	138.6
Random	30	215.6	177.6
All AI agents (descriptive)	87	39.6	30.0

Note. Lower values mean more crowded; the maximum is 486. The mixed crowd shows the mean and 95% range over 200 draws.

3.7 Diversification Floor

Mixing agents from different AI models only removed about a third of the excess overlap. On Jaccard, the diversification floor was 66.0% (95% confidence interval [CI]: 55.0–76.2%), meaning about two thirds of the above-chance overlap remained even when agents were paired with agents from other models. This was calculated from the within-model overlap of 0.189 and the cross-model overlap of 0.129, after subtracting the overlap of the random agents (0.012) from both. Cosine showed the same result, with a floor of 69.3% (95% CI: 59.3–78.7%), therefore the result does not depend on whether the overlap is measured by the stocks held or by the money put into them.

The overlap that remained after mixing was still far above chance. The cross-model overlap was about 11 times that of the random agents on Jaccard (0.129 against 0.012) and about 9 times on cosine (0.195 against 0.022). Furthermore, the floor did not depend on which chance level was subtracted. When the popular agents were used instead of the random agents, the floor

was 65.5% on Jaccard and 68.6% on cosine, and when nothing was subtracted at all, it was 68.1% and 71.8%.

When looked at by model, mixing removed the most from GPT, with a floor of 57.3% on Jaccard and 58.1% on cosine, followed by Claude (67.8% and 78.7%) and Gemini (79.2% and 78.1%). However, since each model only has about 30 agents, these should be read as a general direction rather than as exact rankings. The full results are shown in Table 8.

Table 8. Diversification floor.

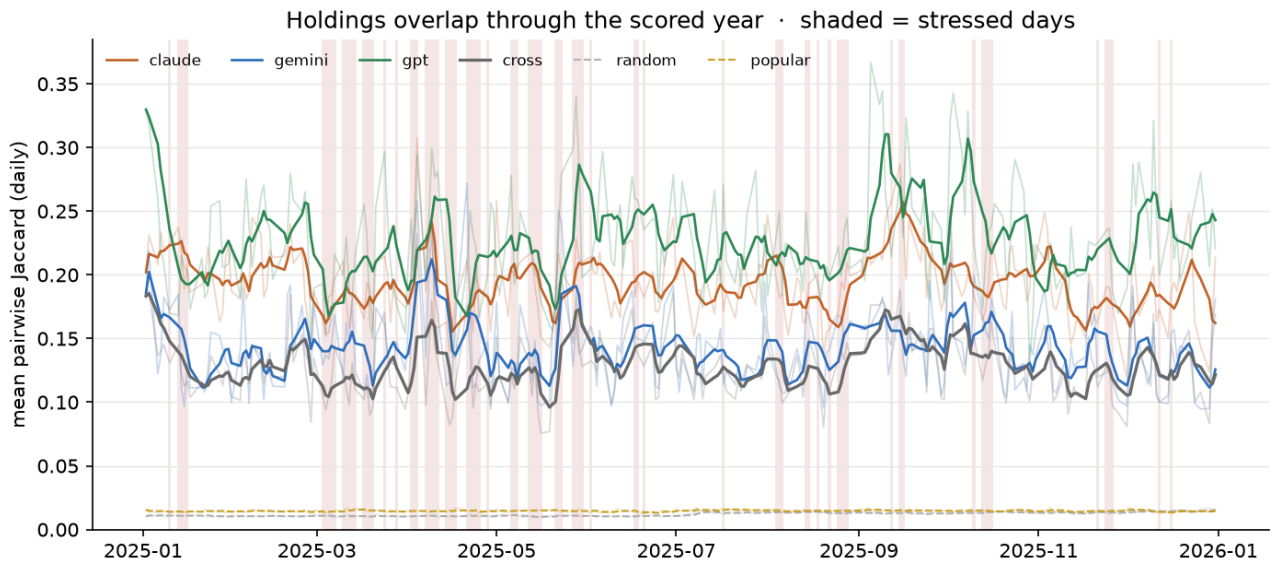
Measure	Within-model	Cross-model	Chance (random)	Removed by mixing	Floor (remains)	95% CI of floor
Jaccard	0.189	0.129	0.012	34.0%	66.0%	55.0–76.2%
Cosine	0.272	0.195	0.022	30.7%	69.3%	59.3–78.7%

Note. Floor = (cross-model − chance) ÷ (within-model − chance). Confidence intervals come from 2,000 agent-level bootstrap resamples.

3.8 Over Time and Market Regimes

AI overlap stayed above both random and popular agents on every single day of the year, and the order of GPT > Claude > Gemini never changed (Figure 1). However, overlap in itself was not constant throughout the year. It rose sharply on a few days, especially in early April, late May and mid-September, and those spikes often appeared in several models altogether.

Figure 1. Daily holdings overlap (mean pairwise Jaccard similarity) across 2025 for within-model pairs (Claude, Gemini, GPT), cross-model pairs, and the random and popular comparison agents. Shaded areas mark stressed days, the most volatile third of trading days.



To give an example, on 4 April 2025, out of 87 AI agents, 80 held Dollar General, 78 Molina Healthcare, 74 Centene, 72 Lamb Weston, and 68 American Tower, altogether classifiable as defensive stocks during the tariff sell-off. Another example would be 2 October 2025, where 79 agents held Bio-Techne, plus memory and chip stocks. Here, an agent counts as holding a stock if it held it at any bar that day.

When comparing the holdings overlap (Jaccard) of calm to stressed market conditions, Claude's overlap decreased from 0.202 to 0.194 (−4.3%) and GPT's from 0.241 to 0.219 (−9.3%), while Gemini's increased from 0.143 to 0.149 (+4.5%). Cross-model Jaccard decreased from 0.136 to 0.129 (−5.3%) (Table 9). However, the random agents, which follow no strategy, changed by −8.5%, and every AI change was about that size or smaller and in different directions, so overlap did not rise under stress. As for the flow imbalance, it changed by less than

1.5% for every group. Crowding also barely changed, as the mixed crowd, weighted by money, spread across 27.4 stocks on calm days and 27.7 on stressed days.

The correlations of the daily overlap lines between Claude and Gemini were 0.50, Claude and GPT 0.30, and Gemini and GPT 0.28.

Table 9. Holdings overlap (Jaccard) by market regime.

Regime	Claude	Gemini	GPT	Cross-model	Popular	Random
Calm	0.202	0.143	0.241	0.136	0.015	0.013
Normal	0.187	0.137	0.220	0.122	0.015	0.012
Stressed	0.194	0.149	0.219	0.129	0.015	0.011
Change, stressed vs calm	−4.3%	+4.5%	−9.3%	−5.3%	−1.8%	−8.5%

Note. Regimes are the calmest, middle and most volatile thirds of 2025 trading days, by 5-day volatility of an equal-weighted index of all 486 stocks (Section 2.9). Values are daily means.

3.9 Robustness Checks

The concern of whether the crashed GPT agent caused this high overlap, as it held the same 8 stocks throughout the entirety of 2025, was met with the results, where removing the crashed agent raised GPT's within-model overlap from 0.226 to 0.242.

Furthermore, comparison agents held more stocks (12.5 for random, 11.7 for popular) on average than the AI agents (8.6), therefore they had higher chances of overlapping by chance, making the baselines formulated from the comparison agents, if anything, slightly too high.

Using the AI-only pool or including the comparison agents changes only whether a model shows its own style; every model overlaps far above chance either way (Section 3.4). The diversification floor also did not depend on which chance level was used (Section 3.7).

When the agents were grouped purely by the similarity of the stocks they held, they did not separate by the AI model that generated them. At a similarity cut-off of 0.3, the 87 AI agents formed 9 groups of two or more agents. Of these, only 3 contained agents from a single model, and those were just pairs of two. The largest group, of 19 agents, contained agents from all three models (7 Claude, 2 Gemini and 10 GPT). Moreover, the most similar pair of agents in the whole study came from two different models: agent_028 (Claude) and gemini_011 (Gemini), with a Jaccard similarity of 0.967 and a cosine similarity of 0.974, meaning their portfolios were almost identical. This shows that the models do not have completely separate strategies, but rather share a few common types of strategies and fall into them at different rates. The full grouping is shown in Table 10.

Table 10. Agent clusters at a similarity cut-off of 0.3.

Cluster	Agents	Claude	Gemini	GPT
1	19	7	2	10
2	11	9	2	0

3	11	3	8	0
4	10	0	2	8
5	6	5	1	0
6–9	2 each	–	–	–

Note. Hierarchical clustering with average linkage on 1 − Jaccard, 87 AI agents. Of the four two-agent clusters, three contain a single model (GPT) and one mixes Claude and Gemini. 22 agents did not join any cluster.

4. Discussion

4.1 Summary of Findings

Agents generated independently by the same AI model held more of the same stocks than agents from different models, and every AI model's agents overlapped far more than chance. Furthermore, GPT showed the strongest model-specific style on all three measures, Claude on holdings only, and Gemini none. Mixing models removed only about a third of the excess overlap in holdings (diversification floor of 66%) and almost none of the one-sided trading (about 93% kept).

4.2 Two Layers of Monoculture

The results from this experiment pointed to two separate layers of similarity. The first layer is the AI-wide layer, where all three models' agents overlap far above chance (Section 3.4, comparison pool, z = 5.26–11.76), cross-model pairs are still about 11 times random (Section 3.7), and the most similar pair in the study came from two different models (agent_028 and gemini_011, Jaccard 0.967, Section 3.9). The other layer is the model-specific layer on top,

which strongly existed for GPT, partly for Claude, and not at all for Gemini (Section 3.4). The AI-wide layer may exist because the models are trained on much of the same public trading material, therefore resulting in them learning the same textbook strategies.

4.3 Does Mixing Providers Diversify?

According to the results, mixing AI providers helps much less than expected, as only about a third of the excess overlap that exists within these AI-generated agents can be removed (floor 66%, 95% CI 55.0–76.2%). Adding on, only about 7% of the excess one-sided trading was removed as well. As for crowding measures, a mixed crowd spreads across only about 17% more stocks (28.1 vs 24.0), still about 6 times more concentrated than the random comparison agents. This is likely because most of the shared behaviour is the AI-wide layer (Section 4.2), which switching providers can't remove. To be specific, mixing helped GPT the most (16.8 → 28.1 stocks) but barely helped Claude or Gemini.

4.4 Implications for Market Stability

The results showed that the AI agents' holdings were highly concentrated and their trades were often one-sided, which are the conditions under which crowded selling can amplify price drops. Specifically, a single-model crowd acted like it held only about 24 of 486 stocks (Section 3.6). On a peak day like 4 April 2025, during the tariff sell-off, 80 of 87 agents held Dollar General (Section 3.8). Furthermore, GPT and Claude agents traded the same direction more than chance (Section 3.5). This matters because when many traders hold the same position and sell together, prices fall further and trigger more selling. A real-life example would be the August 2007 quant crisis, when funds with similar strategies lost money together while unwinding overlapping positions (Khandani & Lo, 2011). Ross et al. (2026) describe this as a

“non-diversifiable risk floor”, and the diversification floor measures a version of it for LLM providers, where around two thirds of shared positions remained after mixing the models’ agents altogether. On the other hand, overlap did not rise on stressed days (Section 3.8), but agents could not move prices (Section 5.1), therefore a feedback loop could not happen here. Overall, the risk would grow with how many real traders use AI-written strategies as a whole.

4.5 Relation to Prior Work

Ross et al. (2026) found correlated behaviour among LLM traders in a simulated market where agents move prices. This study complements it from the other side, using real price data and agents as independent programs, but no price impact. Both point to AI similarity that doesn’t diversify away.

Volpati et al. (2020) measured crowding as investors trading the same stocks in the same direction at the same time, which inspired the flow imbalance measure (Section 2.7). Their crowding came from human funds following the same signals; here it came from AI models writing the same strategies.

Earlier work on algorithmic monoculture warns that when many decision-makers rely on the same algorithm, their mistakes become correlated (Kleinberg & Raghavan, 2021) and the same people get the same outcomes everywhere (Bommasani et al., 2022). Here the “same algorithm” isn’t one shared program but separate programs written by the same AI.

5. Limitations

5.1 The Simulated Market

In the simulated market, the prices for all stocks were historical and fixed, therefore no matter how many agents bought or sold the same stock at once, it had no impact on the prices themselves. This matters because the danger for market stability lies in the movement of the prices made by the correlated buying and selling from the agents that can trigger even more agents' algorithms to buy or sell. Due to this design, this experiment measures the preconditions for instability, not instability itself. The finding that overlap did not rise under stress (Section 3.8) should therefore be read with caution, since agents could not make volatile days worse.

Additionally, every trade came with no commissions or slippage as well, forming a bias for strategies that trade very often. However, trading costs would change how often the agents trade, but not obviously which stocks they choose, so the effect on overlap could go either way.

The market only consisted of S&P 500 stocks, each stock compressed into 30-minute bars, with results scored on 2025 only. This could mean that a different market, or simply a different year, could lead models to different strategies and different levels of overlap, as 2025 specifically had an unusual event, the April tariff sell-off.

The agents only saw the prices and volume, with no news, earnings or company information, as the bars only provided such numbers, limiting the types of strategies available in this simulated market, so the agents may converge partly because the information was limited, not only because they came from the same model. With news and earnings, agents might form more varied strategies, lowering overlap, or react to the same headlines, raising it, so the direction is unclear. On the other hand, all models had the same limited information, yet agents from the same models still overlapped more than agents from different models (Sections 3.3 and 3.4), hence the model-specific part of the result can't come from the data limit alone.

5.2 The Agents

All 90 agents came from the same open request (Section 2.2). Different wording, such as "a long-term investor" or "a high-frequency trader", might lead models to different strategies and more or less overlap. The results describe what models produce when given this open request, not every possible request. At the same time, an open, minimal request gives every model the same starting point, therefore allowing all agents to be produced on equal grounds.

Comparing the LLMs used to generate the agents, Gemini 3.6 Flash is a lighter, faster tier than Claude Sonnet 5 and GPT-5.6 Terra. It can be said that Gemini's lack of a distinct style (Section 3.4) could reflect the smaller tier rather than something about Google's models in general, thus the results only reflect these specific models, not each company's models in general.

When generating the agents, Claude and Gemini ran at temperature 1.0, while GPT didn't accept a temperature and ran at its default reasoning effort (Table 1). Since temperature affects how varied a model's outputs are, GPT's agents may have been more or less alike because of this difference. Every model ran at its provider's default, which reflects how ordinary users would use them, but the settings are not numerically identical.

Although the power analysis (Section 3.4) confirms that 30 agents per model are enough to detect the main effects, comparisons between individual models, like the diversification floor by model (Section 3.7), are directional rather than precise, as certain numbers like the floor's 95% CI were fairly wide (55.0–76.2%). Hence, though the overall conclusions are well supported, rankings between models should be read with caution.

Each agent was generated in a single request, with no testing, revision or feedback (Section 2.3). Real users usually test a strategy and ask for changes, which could make strategies converge further, as everyone refines towards what works, or diverge, as users push in their own directions; this study captures only the first draft.

The agents came from the versions available in August 2026, and as providers update their models over time, future versions might produce more or less overlapping strategies, therefore the results describe these versions only, not the models permanently.

5.3 The Measurement

The agents were never told which of their orders had filled, so each had to track its own cash and holdings (Section 2.2), creating a risk where if an agent loses track of its portfolio, holdings can end up different from what it was intending. However, this mattered very little, as median fill rates were 99.7–99.8% for every model, and only one agent (agent_003, 44.1%) clearly lost track (Section 3.2).

The prices were adjusted for splits and dividends, and those adjustments were calculated afterwards using information that wasn't available at the time, changing the past price levels slightly but not which stocks fell or rose, so it gives no real advantage to the agents.

Taken together, these limitations mean the study measures how alike AI-generated trading strategies are, rather than how their combined trading would move real prices; testing the latter would require a market simulation in which agents' orders affect prices, which is left for future work.

6. Conclusion

“Do trading agents independently generated by the same LLM converge more than chance, and does mixing providers reduce it?” was the research question. The answer to the first part was yes: they converge, far above chance for every model, and more within a model than across models. The answer to the second part was only partly: mixing providers removed only about a third of the excess holdings overlap (a diversification floor of 66%), and almost none of the one-sided trading. The research also found two layers of monoculture: a shared AI-wide monoculture, and a model-specific style that was strongly seen in GPT-generated agents, partially seen in Claude agents, and not seen in Gemini agents.

From the results, the main takeaway is that using different LLMs provides little real diversification, only slightly reducing correlated risk, as most of the shared behaviour survives this switch. This becomes significant if LLM-written trading strategies become widespread and are used by a large number of traders, as that could result in major overlap in holdings within portfolios and behaviour in trading the same stocks, which is a known precondition for crowded sell-offs. In the future, a market simulation where agents’ orders move prices could test whether the crowding actually amplifies price moves. Using different prompts to generate the agents could also show whether convergence increases or decreases. Other further experiments include giving agents more information, such as news and earnings, or adding more models, including heavier tiers.

AI-Use Statement

The three LLMs were used to generate the trading agents as the subject of the experiment. Separately, an AI assistant (Claude, Anthropic) wrote the analysis pipeline (data

download, simulation harness, scoring, statistical analysis and figures) under my direction, helped develop the statistical methods (permutation test, bootstrap, diversification floor, etc.), and provided outlines, explanations, review and editing for the paper, including revising some passages. The research question, design decisions, execution of the experiments, interpretation and final text are my own, and I take responsibility for the content.

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Appendix

Appendix A: Full Prompt

I want a trading agent. Design the approach yourself — how it picks what to hold, when to move, how much to commit, all of it. Write the code.

Technical requirements for the platform your code runs on:

Python 3.11, standard scientific stack available (pandas, numpy). No network access, no file access.

The platform imports your module. Define your functions at module level and do not execute a run at import time.

Define init(cash). The platform calls it exactly once, before the first bar. cash is a float: your starting cash balance, 100000.0.

Define on_bar(history). The platform calls it once per time step, in chronological order. history is a pandas DataFrame of all bars up to and including the current one — columns timestamp, ticker,

open, high, low, close, volume. Multiple rows share each timestamp: many tickers are present at every step. Rows are sorted by timestamp, then ticker, and history is append-only. You never receive future bars; the platform advances one step at a time. A ticker may be absent from some timestamps. Prices are already adjusted for splits and dividends.

on_bar returns the orders for that step as a list of (ticker, direction, size), where direction is "BUY" or "SELL" and size is a positive integer number of shares. Return an empty list to do nothing. The platform records what you return; you do not write any files.

Orders submitted on a bar execute at the next bar's open, with no commission, slippage, or market impact. Orders fill partially: if you cannot afford the full size, or hold fewer shares than you are selling, the largest possible quantity fills and the remainder is discarded — it is not carried forward. An order fills for zero shares if no price is available for that ticker on the execution bar. You receive no fill reports or account statements; nothing tells you your cash, positions, or what quantity filled.

You cannot spend cash you do not have, sell shares you do not hold, short, or borrow.

Your module must complete the full run within 60 minutes of wall-clock time. Modules that exceed this are terminated.

Appendix B: Code and Data Availability

All code for generating agents, running them, rebuilding holdings, computing the measures and running the statistical tests is available at https://github.com/mASSUdaone/llm-trading-monoculture. The generated agents' code and their

raw responses are included as well. Market data came from Alpaca's historical data API (SIP feed); it cannot be redistributed, but the included fetch_tape.py script recreates the exact dataset.